

Fibonacci Acoustic Scanner Pro

Complete User Guide · v1.0

Dynamic Occlusion · Diffraction · Real-time Reverb

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Unity 2021.3+ · Wwise 2022.1+ · Wwise Unity Integration required

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1. Overview

Fibonacci Acoustic Scanner Pro is a runtime acoustic simulation plugin for Unity + Wwise. It uses a Fibonacci sphere of raycasts to model how sound behaves in real environments -- attenuating through walls, diffracting around corners, and reverberating according to the size and surface materials of the space the listener occupies.

Feature	Description
Occlusion	Multi-ray sampling with wall thickness estimation
Diffraction	Edge-based diffraction around obstacles
Room Scanning	Fibonacci sphere drives reverb decay and send level
Material System	Per-tag absorption coefficients with debug colours
LOD System	Update rate scales with distance to listener
Editor Tools	One-click scene setup, collider fixer, RTPC monitor

2. Installation

Unity

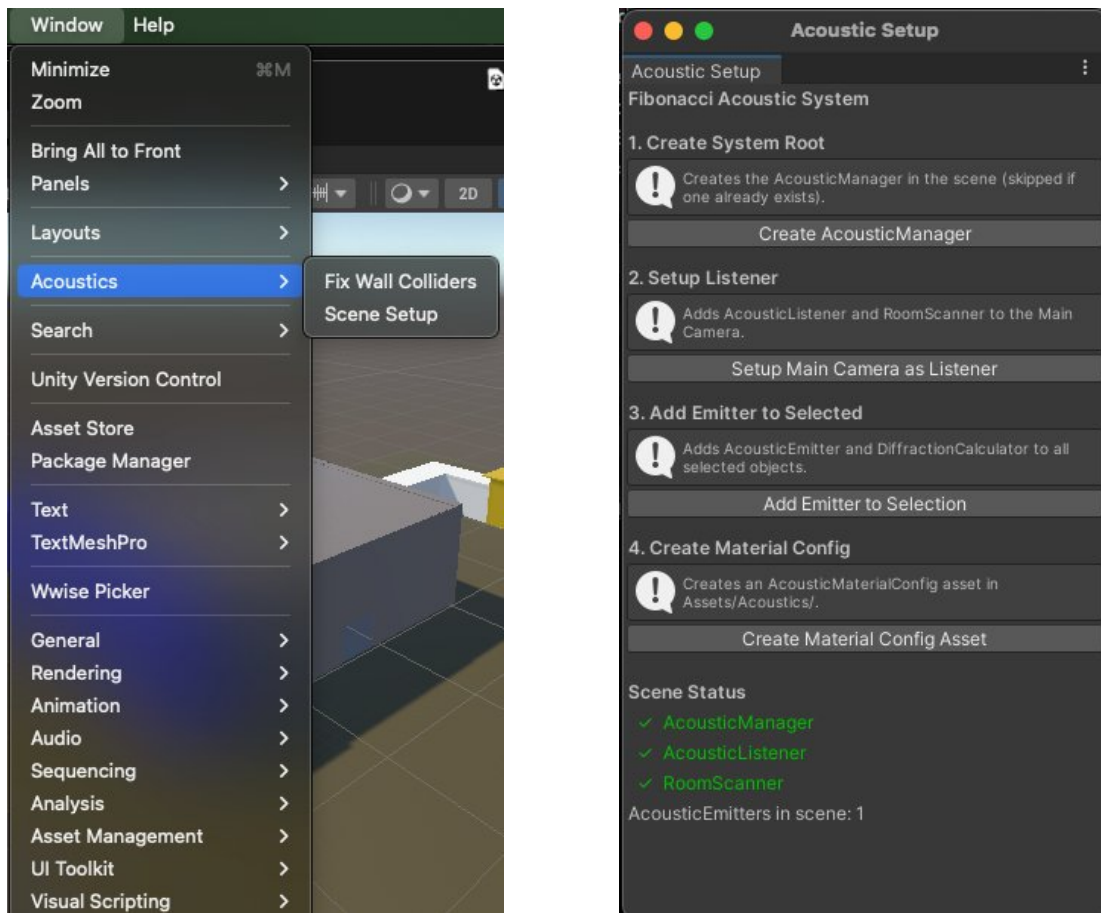
1. Import **FibonacciAcousticScannerPro_v1.0.unitypackage** into your project.
2. Ensure Wwise Unity Integration is installed and **AK_WWISE_UNITY_API** define is present.
3. Open **Window > Acoustics > Scene Setup** and follow the four steps.

Wwise

1. Install Python 3.7+ on your machine.
2. Enable WAAPI: **Project > User Preferences > Enable Wwise Authoring API**.
3. Navigate to the **Wwise/** folder and run: **python setup_wwise.py**
4. The script creates all required Game Parameters, Aux Bus, and RTPC curves.
5. Assign the generated Game Parameters to **AcousticRTPCConfig** in Unity.
6. Generate SoundBanks: **Generate All** in SoundBank Manager.

3. Quick Start -- Scene Setup

Open the **Scene Setup** window via **Window > Acoustics > Scene Setup**.



Left: Window > Acoustics menu · Right: Scene Setup window

Step 1 -- Create AcousticManager

Creates `_AcousticSystem` GameObject with `AcousticManager`. Skipped if already exists.

Step 2 -- Setup Main Camera as Listener

Adds `AcousticListener` and `RoomScanner` to the Main Camera.

Step 3 -- Add Emitter to Selection

Select objects in the Hierarchy, click to add `AcousticEmitter` and `DiffractionCalculator`.

Step 4 -- Create Material Config Asset

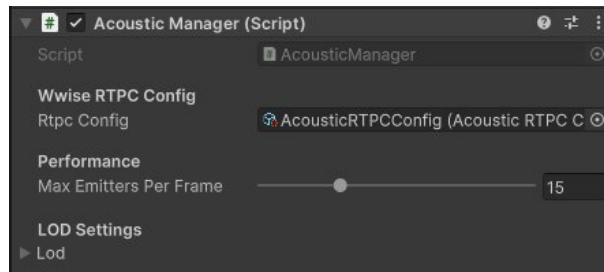
Creates `AcousticMaterialConfig` in `Assets/Acoustics/`. Assign to `RoomScanner` and `AcousticEmitter`.

Scene Status shows green checkmarks when all three required components are present.

4. Core Components

4.1 AcousticManager

Central coordinator. Attach to `_AcousticSystem` GameObject.



AcousticManager Inspector

Rtpc Config

Assign AcousticRTPCConfig ScriptableObject. Required for Wwise communication.

Max Emitters Per Frame

Maximum emitters updated per frame. Default: 15.

LOD -- Close/Medium Distance

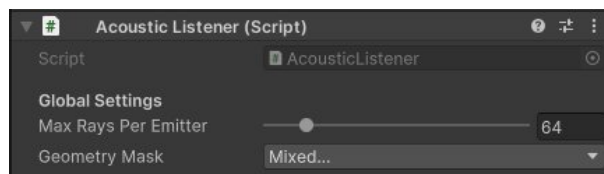
Distance thresholds for LOD tier switching.

LOD -- Update Rate Close/Medium/Far

Frames skipped between updates. 1 = every frame, 4 = every 4th frame.

4.2 AcousticListener

Attach to same GameObject as Unity AudioListener (player head or Main Camera).



AcousticListener Inspector

Max Rays Per Emitter

Maximum rays per emitter per scan. Default: 64.

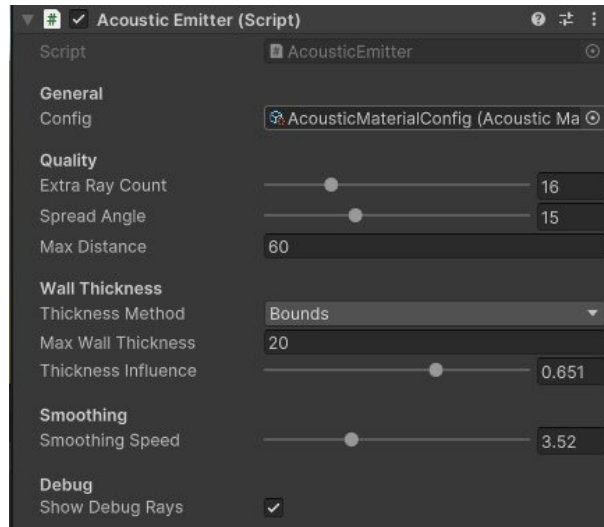
Geometry Mask

Layers blocking sound. **Exclude Player and Pickable layers.**

Note: Always exclude the Player layer from Geometry Mask.

4.3 AcousticEmitter

Attach to any sound-emitting GameObject. Sends occlusion data to Wwise every frame.



AcousticEmitter Inspector

General

Config

Assign an AcousticMaterialConfig asset.

Quality

Extra Ray Count

Additional rays in cone around primary ray. Range: 1-64. Default: 16.

Spread Angle

Half-angle of ray cone in degrees. Range: 0-45. Default: 15.

Max Distance

Beyond this the emitter reports zero occlusion. Default: 60.

Wall Thickness

Thickness Method

None = no estimation. **Bounds** = fast bounding box (recommended). **DoubleRay** = accurate but expensive.

Max Wall Thickness

Maximum thickness in metres. Default: 5.

Thickness Influence

Thickness effect on occlusion. 0-1. Default: 0.4.

Smoothing

Smoothing Speed

Lerp speed toward target occlusion. Range: 0.5-10. Default: 3.

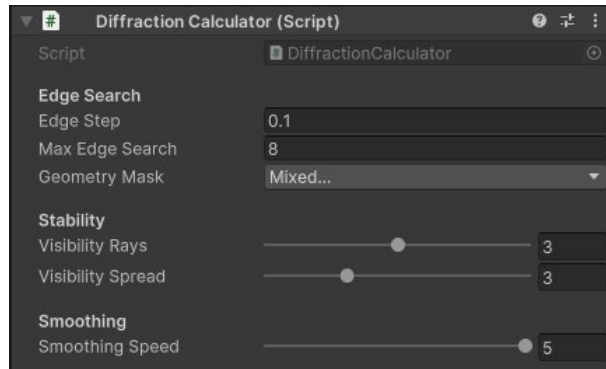
Debug

Show Debug Rays

Green = sky, Red = geometry, Magenta = thickness probe. **Disable in builds.**

4.4 DiffractionCalculator

Calculates how much sound bends around obstacles.



DiffractionCalculator Inspector

Edge Search

Edge Step

Step size when scanning edges. Default: 0.1.

Max Edge Search

Maximum search radius in metres. Default: 8.

Geometry Mask

Should match AcousticListener Geometry Mask.

Stability

Visibility Rays

Rays for line-of-sight confirmation. Range: 1-5. Default: 3.

Visibility Spread

Angular spread in degrees. Default: 3.

Smoothing

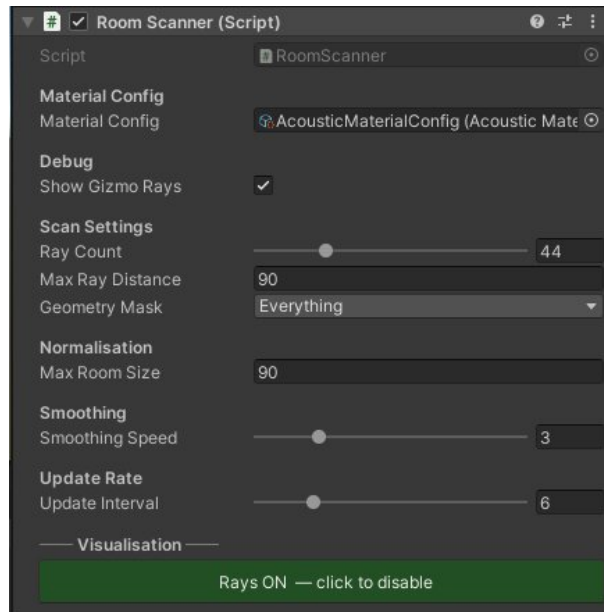
Smoothing Speed

Range: 0.5-5. Default: 1.5. Increase to 8-12 to fix sharp jumps along walls.

Note: Sharp diffraction jumps along walls outdoors: increase Smoothing Speed to 8-12.

4.5 RoomScanner

Scans the space around the listener to derive room size and reverb amount.



RoomScanner Inspector

Material Config

Assign AcousticMaterialConfig for per-surface absorption.

Show Gizmo Rays

Draws scan rays in Scene view. Disable in builds.

Ray Count

Fibonacci sphere density. Range: 16-128. Default: 44.

Max Ray Distance

Maximum ray travel distance. Default: 90.

Geometry Mask

Layers included in scanning. Typically Everything.

Max Room Size

Distance at which RoomSize reaches maximum. Default: 90.

Smoothing Speed

Lerp speed for RoomSize and ReverbAmount. Default: 3.

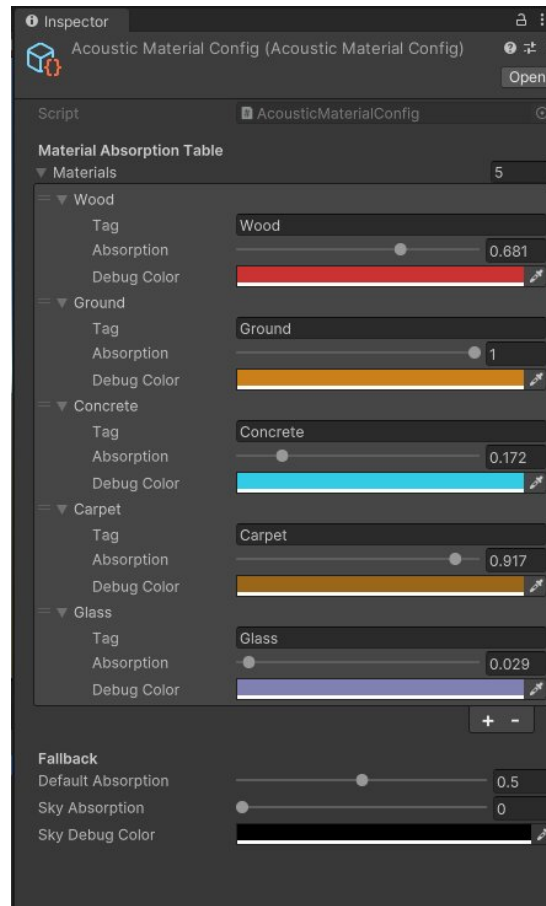
Update Interval

Scan every N frames. 0 = every frame. Default: 6.

5. Configuration Assets

5.1 AcousticMaterialConfig

Maps Unity tags to absorption coefficients. Create via **Assets > Create > Acoustics > Material Config**. Assign to both RoomScanner and AcousticEmitter.



AcousticMaterialConfig -- material table and fallback settings

Material Table

Tag	Absorption	Meaning
Wood	0.681	Medium-high -- wood panels, floors
Ground	1.0	Maximum -- absorbs almost all sound
Concrete	0.172	Low -- highly reflective surface
Carpet	0.917	Very high -- soft furnishings
Glass	0.029	Very low -- nearly fully reflective

Fallback Settings

Default Absorption

Used when no matching tag found. Default: 0.5.

Sky Absorption

Applied to rays missing all geometry. Set 0.9 to suppress outdoor reverb.

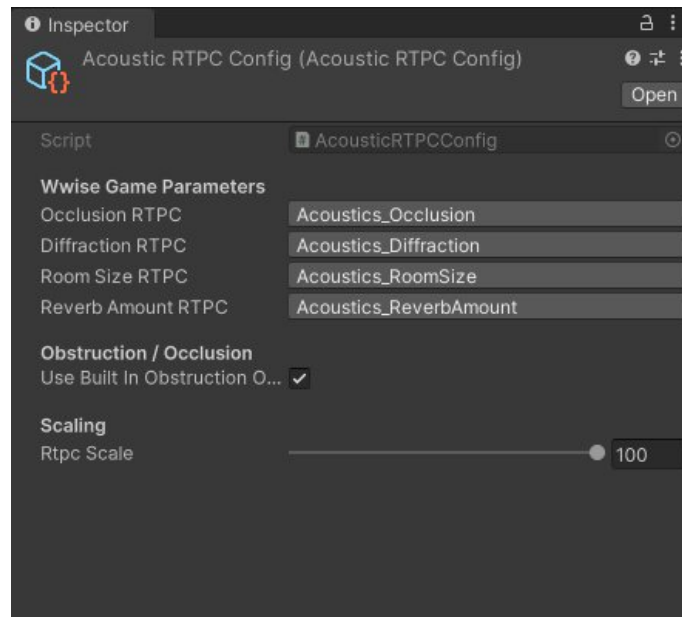
Sky Debug Color

Colour of sky rays in Scene view.

Note: Absorption 0 = fully reflective, 1 = fully absorptive.

5.2 AcousticRTPCConfig

Links Unity acoustic system to Wwise Game Parameters. Create via **Assets > Create > Acoustics > RTPC Config**. Assign to **AcousticManager > Rtpc Config**.



AcousticRTPCConfig -- Wwise Game Parameter assignments

Occlusion RTPC

Receives occlusion (0-100). Controls Voice Volume and LPF.

Diffraction RTPC

Receives diffraction (0-100).

Room Size RTPC

Receives room size (0-100). Drives RoomVerb Decay Time.

Reverb Amount RTPC

Receives reverb amount (0-100). Controls Aux Bus volume. Silent 0-35, audible 35-100.

Use Built-in Obstruction Occlusion

Also uses Wwise native per-object obstruction/occlusion.

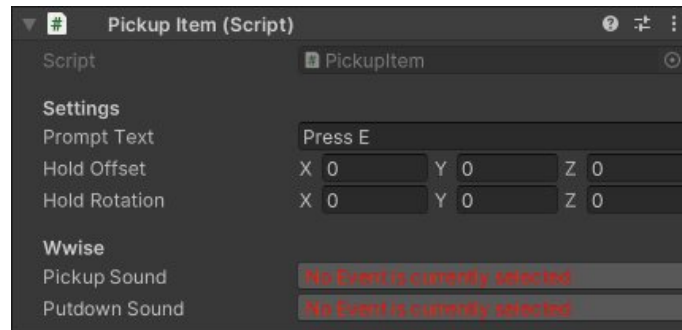
RTPC Scale

Multiplier before sending to Wwise. Default: 100.

6. Demo Components

6.1 PickupItem

Attach to any prop the player can carry. Works with **PlayerPickup**.



PickupItem Inspector

Prompt Text

Text shown when player is in pickup range.

Hold Offset

Position offset relative to hold point when carried.

Hold Rotation

Euler rotation applied while held.

Pickup Sound

Wwise Event posted on pickup.

Putdown Sound

Wwise Event posted on putdown.

Note: AcousticEmitter stays active while carried -- reverb updates in real time between rooms.

PlayerPickup

Attach to the player. Handles pick up / put down with a configurable key.

Pickup Key

Key to pick up or drop. Default: E.

Hold Point

Transform the held item is parented to.

Pickup Radius

Detection radius for pickable items.

Pickup Mask

Layer mask for PickupItem search. Set to Pickable layer.

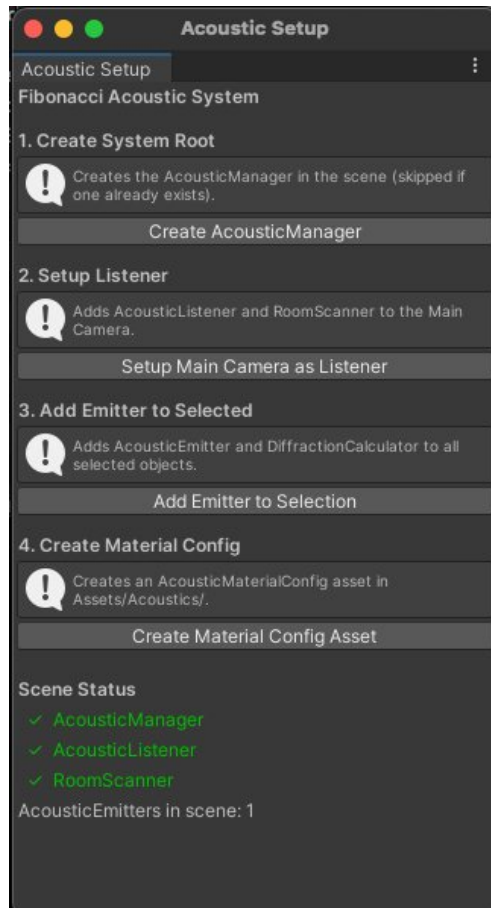
Show Prompt

Displays pickup prompt on screen.

7. Editor Tools

7.1 Scene Setup Window

Open via **Window > Acoustics > Scene Setup**.

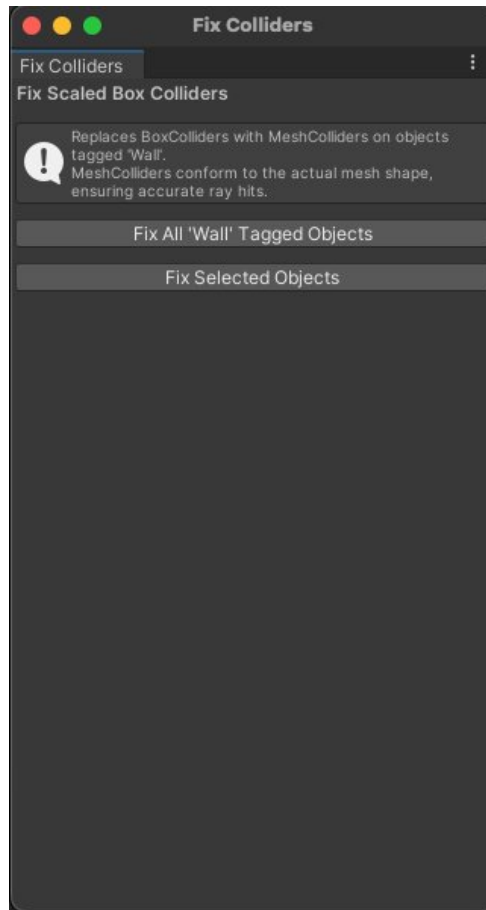


Scene Setup window

Scene Status shows green checkmarks when AcousticManager, AcousticListener, and RoomScanner are present. Emitter count shows how many AcousticEmitters are in the scene.

7.2 Fix Wall Colliders

Open via **Window > Acoustics > Fix Wall Colliders**.



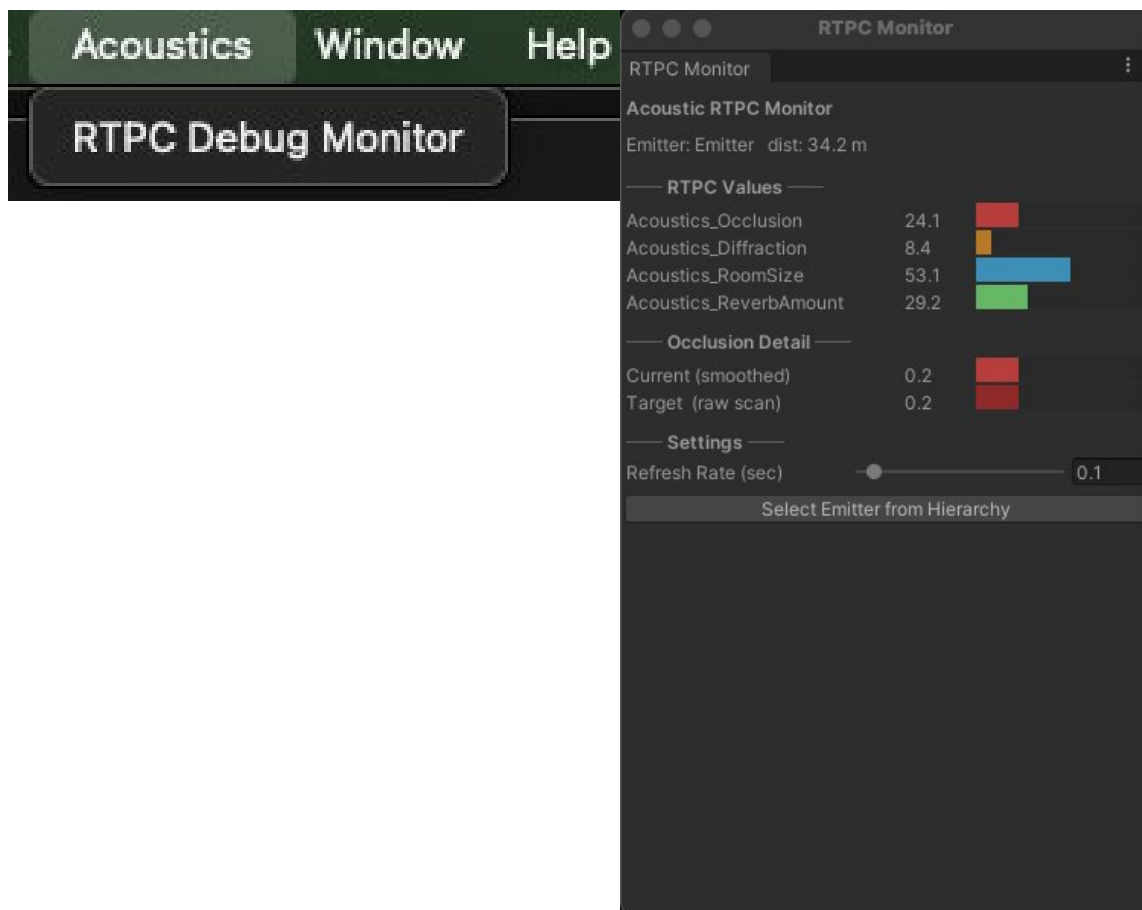
Fix Wall Colliders window

Fix All Wall Tagged Objects -- processes all GameObjects tagged "Wall". **Fix Selected Objects** -- processes selected objects only. BoxColliders on scaled geometry do not match the actual mesh, causing rays to miss walls.

Note: Use MeshColliders on all wall geometry for accurate occlusion results.

7.3 RTPC Debug Monitor

Open via **Acoustics > RTPC Debug Monitor** in the menu bar.



Left: Acoustics menu · Right: RTPC Debug Monitor window

RTPC	Range	What it drives
Acoustics_Occlusion	0-100	Voice Volume + Low-pass filter (per emitter)
Acoustics_Diffraction	0-100	Voice Volume + Low-pass filter (global)
Acoustics_RoomSize	0-100	RoomVerb Decay Time (0.2s - 5.0s)
Acoustics_ReverbAmount	0-100	Aux Bus Volume (silent 0-35, audible 35-100)

Use **Select Emitter from Hierarchy** to switch to a specific emitter. **Refresh Rate** slider controls update frequency (default 0.1s).

8. Wwise Setup

Automatic Setup (Recommended)

Run the included Python script from the **Wwise/** folder.

```
pip install waapi-client python setup_wwise.py
```

Optional: **--sound "MySound"** targets a specific sound, **--sound-path "\\path"** specifies its Wwise location, **--no-sound** skips sound configuration.

Manual Setup Summary

Step	Action
1	Create 4 Game Parameters: Acoustics_Occlusion, Acoustics_Diffraction, Acoustics_RoomSize, Acoustics_ReverbAmount (range 0-100)
2	Create Aux Bus "Aux_Reverb" under Master Audio Bus
3	Add Wwise RoomVerb effect to Aux_Reverb
4	RTPC on Aux_Reverb: Output Bus Volume -> Acoustics_ReverbAmount (0 = -200 dB, 35 = -200 dB, 100 = 0 dB)
5	RTPC on RoomVerb: Decay Time -> Acoustics_RoomSize (0 = 0.2s, 100 = 5.0s)
6	On each sound: enable Game-Defined Aux Sends, set Early Reflections -> Aux_Reverb
7	On each sound: add RTPC curves for Occlusion, Diffraction, RoomSize, ReverbAmount
8	Generate All SoundBanks, assign Game Parameters to AcousticRTPCConfig in Unity

9. Performance Tips

Disable debug rays in builds

Set Show Debug Rays = false on AcousticEmitter and Show Gizmo Rays = false on RoomScanner.

Tune Max Emitters Per Frame

Reduce to 5-8 on mobile. LOD ensures distant emitters still update.

Increase Update Interval

Set RoomScanner Update Interval to 8-12 for large open scenes.

Use Bounds thickness method

DoubleRay is 2x more expensive. Use only when precision is critical.

Reduce Extra Ray Count outdoors

8 rays sufficient outdoors. Reserve 16+ for interior-heavy scenes.

Geometry Mask

Exclude layers that never block sound (water, foliage, triggers).

10. Troubleshooting

No occlusion / reverb at all

Check AcousticManager has AcousticRTPCConfig. Verify all 4 Game Parameters are set. Regenerate SoundBanks after Wwise changes.

Occlusion = 100% next to emitter

Remove Player layer from AcousticListener Geometry Mask.

Reverb not audible

Check Aux_Reverb Output Bus. Verify "Use game-defined aux sends" is on. Check User-Defined Aux Sends slot 0 has Aux_Reverb with Volume > -200.

Reverb too loud outdoors

Increase Sky Absorption to 0.9. Check ReverbAmount RTPC curve has flat -200dB section for values 0-35.

Reverb does not change when carrying emitter

Fixed in v1.0 -- AcousticEmitter sends ReverbAmount every frame. Ensure latest AcousticEmitter.cs.

Sharp diffraction jump along wall

Increase DiffractionCalculator Smoothing Speed to 8-12. Try Edge Step 0.3.

Emitter falls through floor

Need physics collider (Is Trigger=false) AND trigger collider (Is Trigger=true). Exclude Pickable from Geometry Mask.

AcousticManager not found

_AcousticSystem must exist before emitters register. Use DontDestroyOnLoad or check scene load order.

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For support, feedback, or feature requests, contact K.K.Sound.